

Python Programming Test

Topics: for Loop • if / elif / else • Nested Loops • Comprehensions

Instructions: Attempt all questions. Write clean Python code. Do not use built-in shortcuts unless the question permits them.

Section A — if / elif / else

1. Write a program to input a number and check whether it is positive, negative, or zero using if, elif, and else.
2. Input a student's marks and print Grade A for 80–100, Grade B for 60–79, Grade C for 40–59, and Fail for below 40.
3. Write a program to check whether a given year is a leap year.
4. Input three numbers and find the largest number using if / elif / else.
5. Input a person's age and classify them as Child, Teenager, Adult, or Senior Citizen.
6. Check whether a number is divisible by 3, 5, both 3 and 5, or neither.
7. Create a simple calculator using if / elif / else for +, -, *, and /.
8. Input a character and determine whether it is uppercase, lowercase, a digit, or a special character.

Section B — for Loop

9. Print all numbers from 1 to 50 using a for loop.
10. Print all even numbers between 1 and 100 using a for loop.
11. Print all odd numbers between 1 and 100 using a for loop.
12. Calculate the sum of numbers from 1 to 100 using a for loop.
13. Print the multiplication table of a number entered by the user.
14. Find the factorial of a given number using a for loop.
15. Count how many vowels are present in a given string using a for loop.
16. Find the largest number in a list without using max().
17. Count the number of positive, negative, and zero values in a list.
18. Reverse a string using a for loop without using slicing [::-1].

Section C — for Loop with if / elif / else

19. Print numbers from 1 to 100, but print Fizz for multiples of 3, Buzz for multiples of 5, and FizzBuzz for multiples of both.
20. From a list of numbers, print only numbers that are divisible by both 2 and 3.
21. Given a list of marks, print the grade of every student using if / elif / else.
22. Given a list of numbers, count how many are even and how many are odd.

23. Print all prime numbers between 1 and 100 using a for loop and conditional statements.
24. Given a list of numbers, calculate the sum of only positive numbers.
25. Given a list of numbers, print Positive Even, Positive Odd, Negative Even, or Negative Odd for each number.

Section D — Nested for Loops

26. Print this pattern using nested loops: * / ** / *** / **** / *****.
27. Print a 5 × 5 square pattern of stars using nested loops.
28. Print multiplication tables from 1 to 10 using nested for loops.
29. Print all pairs (i, j) where i and j range from 1 to 5.
30. Given a matrix, print every element using nested for loops.
31. Find the sum of all elements in a 2D list using nested for loops.

Section E — List Comprehension

32. Create a list containing the squares of numbers from 1 to 20 using list comprehension.
33. From a list, create a new list containing only even numbers using list comprehension.
34. From a list, create a new list containing only numbers greater than 50.
35. Convert all strings in a list to uppercase using list comprehension.
36. Given a list of words, create a list containing the length of every word.
37. Extract all vowels from a string using list comprehension.
38. Create a list of numbers from 1 to 100 that are divisible by both 3 and 5.
39. Flatten [[1, 2], [3, 4], [5, 6]] using nested list comprehension.

Section F — Dictionary & Set Comprehension

40. Create a dictionary from 1 to 10 where each key is a number and each value is its square.
41. Given a student dictionary containing names and marks, create a new dictionary containing only students who scored 50 or more.
42. Given a list of words, create a dictionary where each word is the key and its length is the value.
43. Create a dictionary from 1 to 20 mapping each number to Even or Odd.
44. Given a list of numbers, create a set containing their squares using set comprehension.
45. Extract all unique vowels from a string using set comprehension.

Section G — Challenge Questions

46. Find the second-largest number in a list without using sort() or sorted().
47. Find all duplicate elements in a list using loops and conditional statements.
48. Check whether a number is an Armstrong number.

49. Print this number pattern using nested loops: 1 / 12 / 123 / 1234 / 12345.

50. Given a list of integers, create a dictionary with keys positive, negative, and zero, where each value contains the corresponding numbers. Use loops and conditional statements.

Suggested Duration: 90–120 minutes **Total Questions:** 50